Unit 3: Biology of the mind

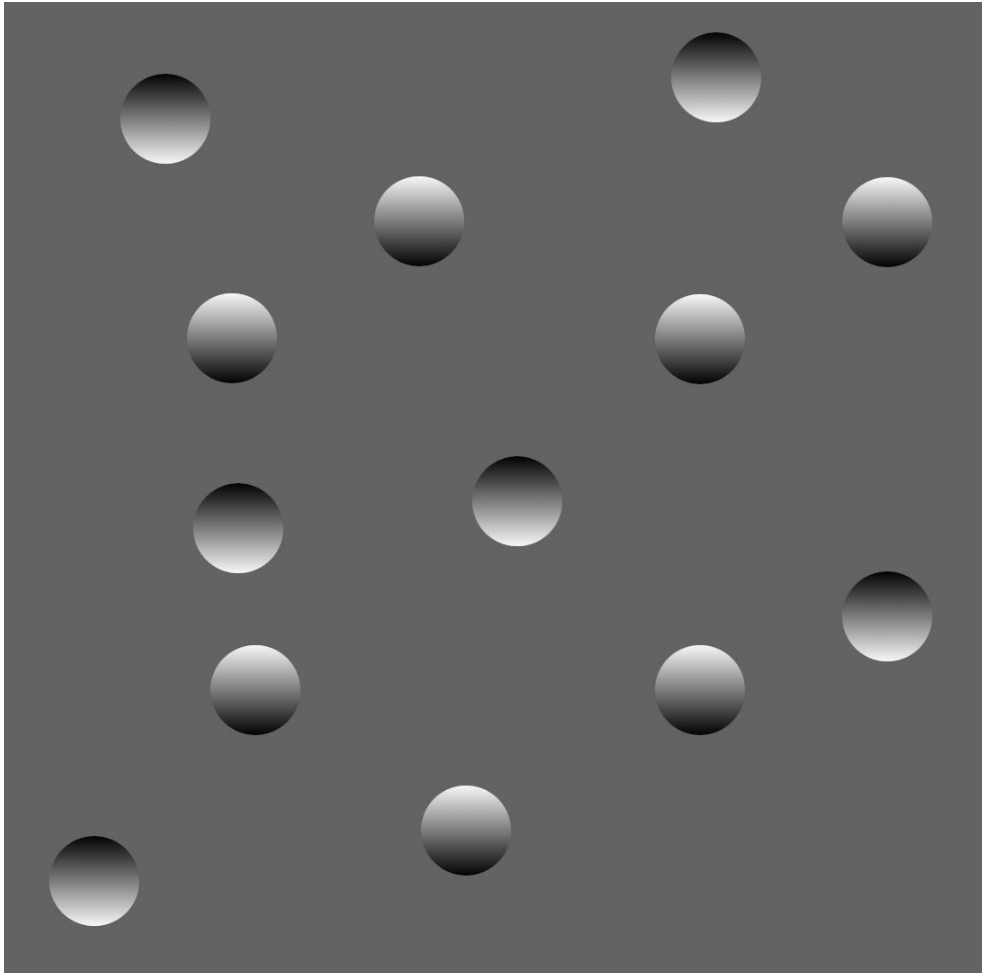
Evolved brain:

The primate brain appeared between 34 million and 23 million years ago, during the Oligocene epoch.

It evolved into the progressively larger brains of the great apes in the Miocene epoch between roughly 23 million and 7 million years ago.

The human lineage diverged from the last common ancestor that we shared with the chimpanzee somewhere in the range of 5 million to 7 million years ago.

Since that divergence, our brains have evolved into the present human brain.

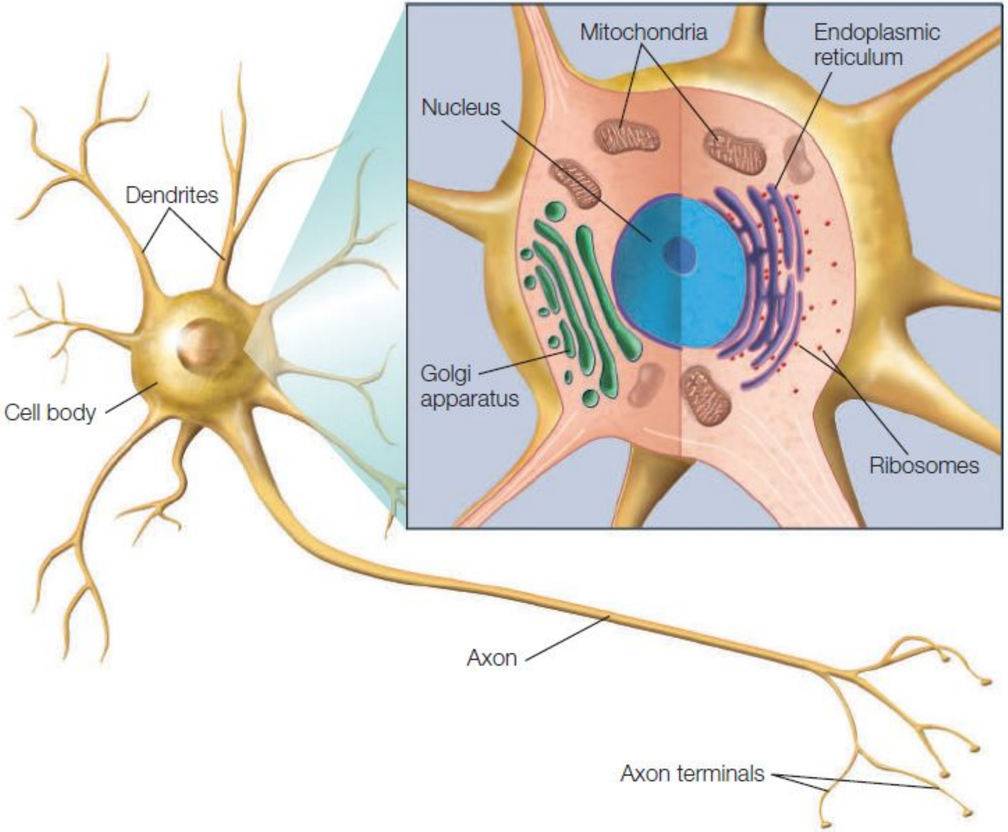


The cells of the nervous system

Neurons are the basic signaling units that transmit information throughout the nervous system.

Glial cells serve various functions in the nervous system, providing structural support and electrical insulation to neurons, modulating neuronal activity, cleaning up cellular debris, forming the blood-brain barrier.

Neurons


Mitochondria
Endoplasmic reticulum
Nucleus
Dendrites
Golgi apparatus
Cell body
Ribosomes
Axon
Axon terminals

Central nervous system: Brain and spinal cord. Command-and-control center of the nervous system.

Peripheral nervous system: nerves (bundles of axons and glial cells) and ganglia (clumps of nerve cell bodies) outside of the CNS. The PNS represents a courier network that delivers sensory information to the CNS and carries motor commands from the CNS to the muscles.

Somatic motor system: controls the voluntary muscles of the body**,**

Autonomic motor system: controls the automated visceral functions

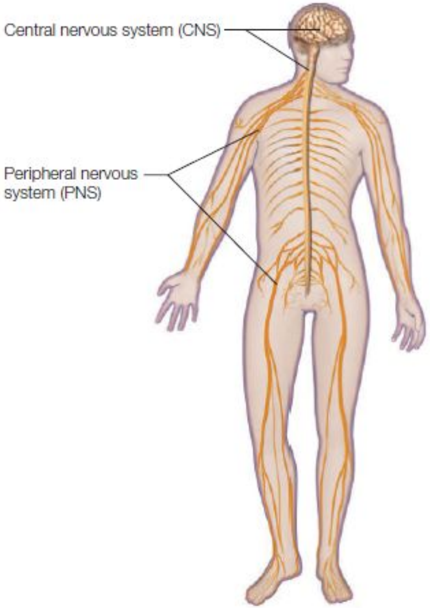


FIGURE 2.17 The peripheral and central nervous systems of the human body.
The nervous system is generally divided into two main parts. The central nervous system (CNS) includes the brain and spinal cord. The peripheral nervous system (PNS), comprising the sensory and motor nerves and associated nerve cell ganglia (groups of neuronal cell bodies), is located outside the central nervous system.

The autonomic nervous system

Regulates the body's internal world.

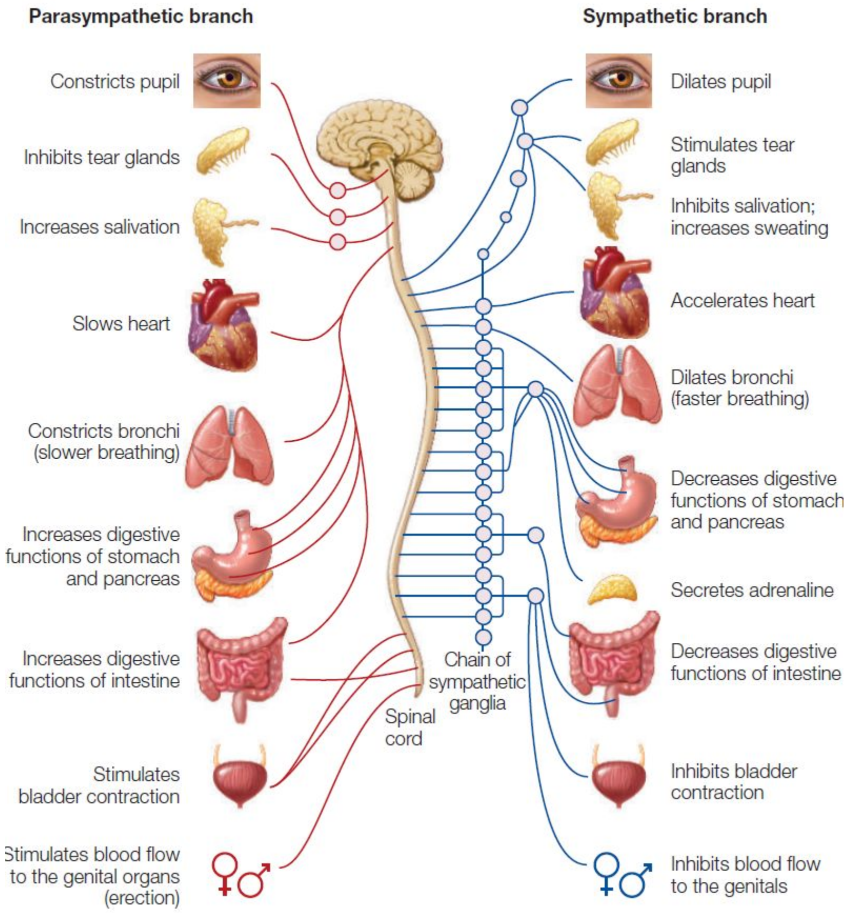
Involved in controlling the involuntary action of smooth muscles, the heart, and various glands.

It has two subdivisions: the sympathetic and parasympathetic branches.

The sympathetic system uses the neurotransmitter norepinephrine, and the parasympathetic system uses acetylcholine as its transmitter.

Activation of the sympathetic system increases heart rate, diverts blood from the digestive tract to the somatic musculature, and prepares the body for action (fight or flight) by stimulating the adrenal glands to release adrenaline.

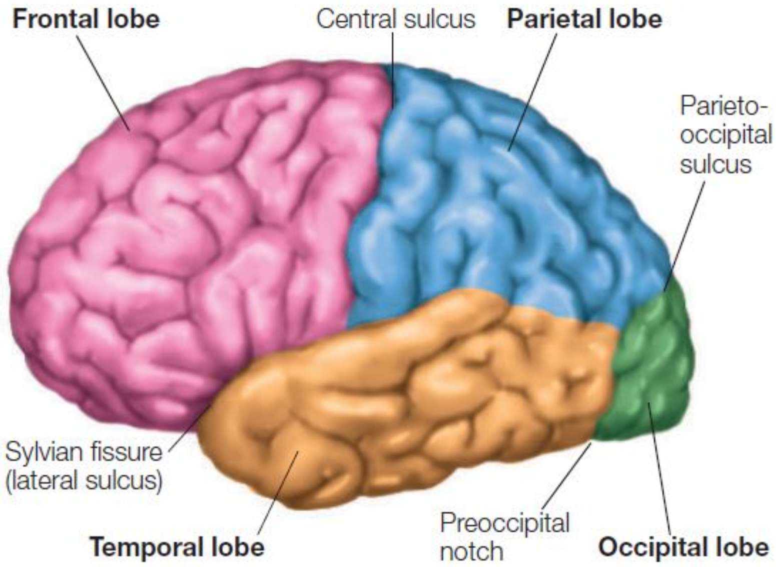
In contrast, activation of the parasympathetic system slows heart rate, stimulates digestion, and in general helps the body with functions germane to maintaining itself (rest and digest).



Divisions of the cerebral cortex

The cortex of each hemisphere has four main divisions: frontal, parietal, temporal, and occipital lobes.

The lobes can usually be distinguished from one another by prominent anatomical landmarks such as pronounced sulci.



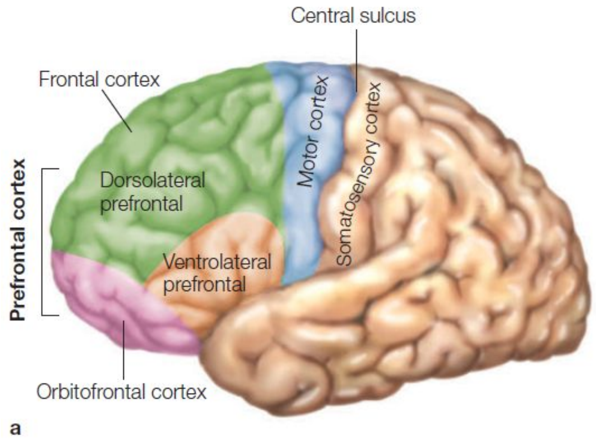
Frontal Lobe: Motor Cortex

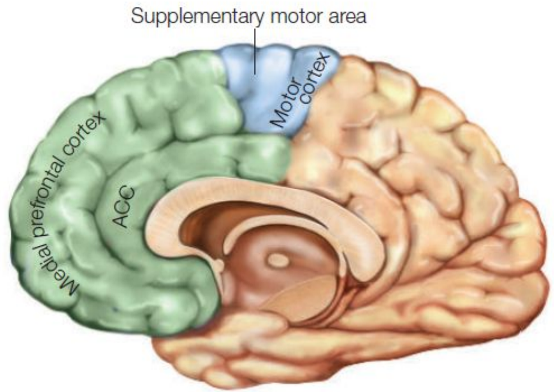
The frontal lobe has two main functional subdivisions: the prefrontal cortex and the **motor cortex**.

The motor cortex sits in front of the central sulcus, beginning in the depths of the sulcus and extending anteriorly.

The **primary motor cortex** is mainly responsible for **generating neural signals that control movement**.

Supplementary motor cortex is involved in the **planning and sequencing** of movement.





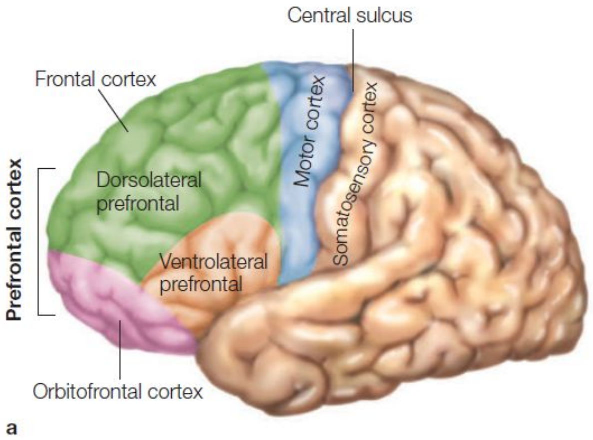
Frontal Lobe: Prefrontal cortex

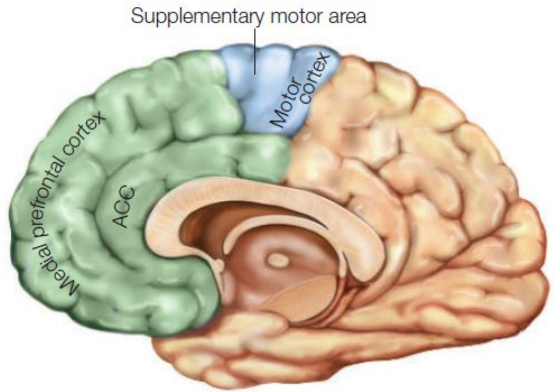
The last to develop and is the evolutionarily youngest region of the brain.

The more complex aspects of planning, organizing, controlling, and executing behavior—tasks that require the integration of information over time.

Center of **cognitive control**.

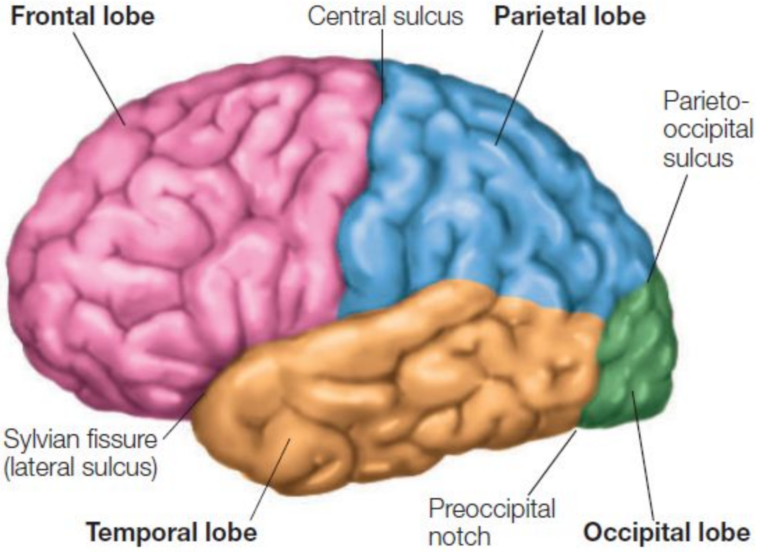
Lesions leads to difficulty in reaching a goal, lack of motivation to initiate action, to modulate it, or to stop it once it is happening.

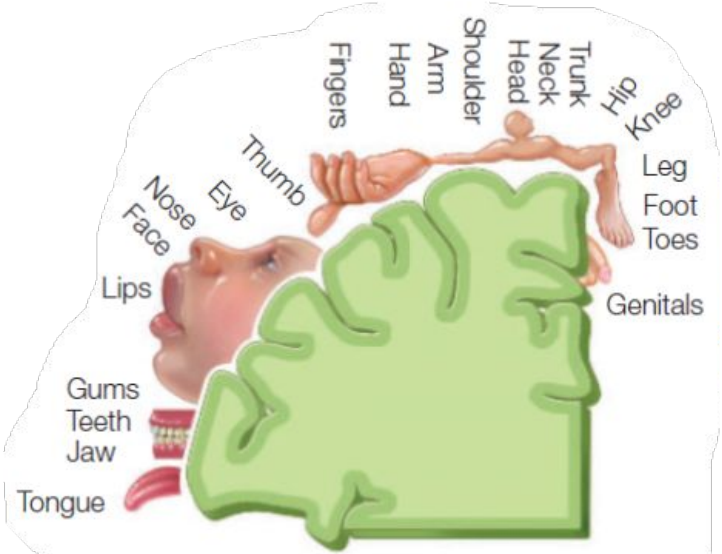




Parietal Lobe: Somatosensory Cortex

The parietal lobe receives sensory information about touch, pain, temperature sense, and limb proprioception (limb position).





Somatosensory cortex

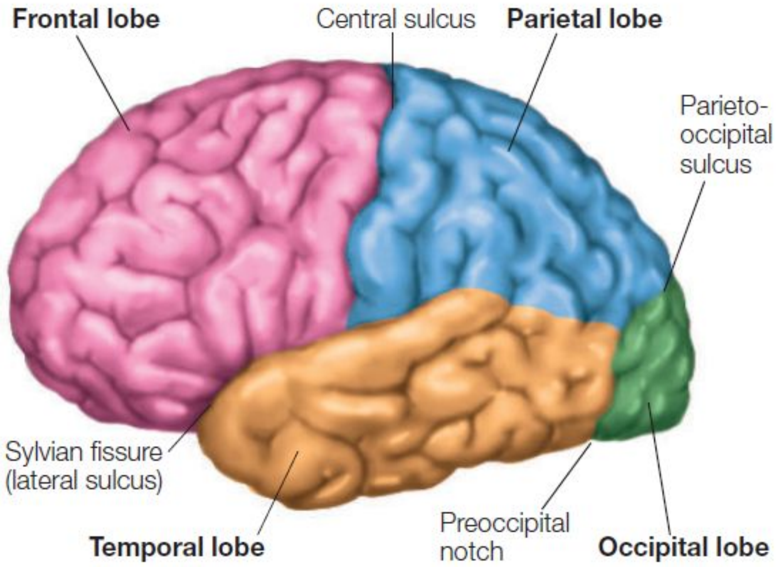
People with lesions may think that parts of their body are not their own or that parts of space don’t exist for them;

they may recognize objects only from certain viewpoints, or be unable to locate objects in space at all.

Stimulating certain regions of the parietal lobe causes people to have “out-of-body” experiences.

Occipital Lobe: Visual Processing Areas

- The occipital lobes at the posterior part of the cerebral cortex are responsible for visual processing.
- Damage to this area often produces a “hole” in the person’s field of vision: Objects in a particular location can’t be seen, but the rest of the visual field may remain unaffected.
- Damage to the occipital lobe in the right hemisphere produces loss of vision in the left visual field, whereas damage to the occipital lobe in the left hemisphere produces loss of vision in the right visual field.

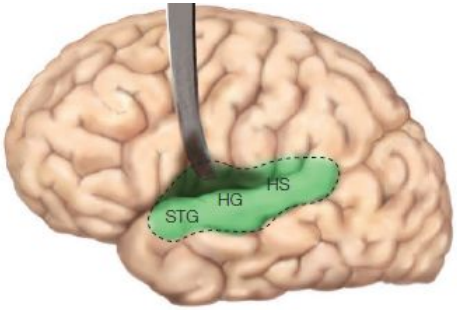


Temporal Lobe: Auditory processing areas

Primary auditory cortex lies in the superior part of the temporal lobe.

When injuries occur in the left hemisphere, people may lose the ability to understand spoken words.

When damage is restricted to the right hemisphere, they may be able to recognize speech but may lose the ability to recognize other organizations of sound—for example, melodies, tones, or rhythms.



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Areas of the cortex that either control motor movements (motor cortex) or receive sensory input (sensory cortex) account for only 20 to 25 percent of the total area.

The remainder is known as the **association cortex**.

These have role in integrating the activities in the various sensory systems and in translating sensory input into programs for motor output.

Also involved in complex cognitive activities such as thinking, reasoning, remembering, language, recognizing faces, and a host of other complex cognitive functions.

Hemispheric Specialization

Wada test

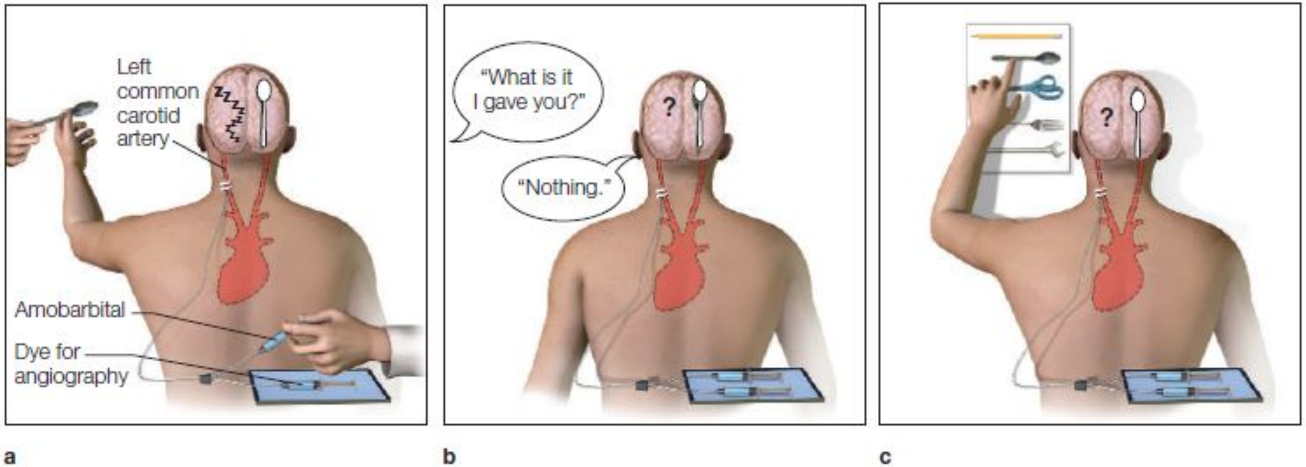


FIGURE 4.2 Methods used in amobarbital testing.
(a) After angiography, amobarbital is administered to the left hemisphere, anesthetizing the language and speech systems. A spoon is placed in the left hand, and the right hemisphere takes note. (b) When the left hemisphere regains consciousness, the participant is asked what was placed in his left hand, and he responds, "Nothing." (c) Showing the patient a board with a variety of objects pinned to it reveals that the patient can easily point to the appropriate object, because the right hemisphere directs the left hand during the match-to-sample task.

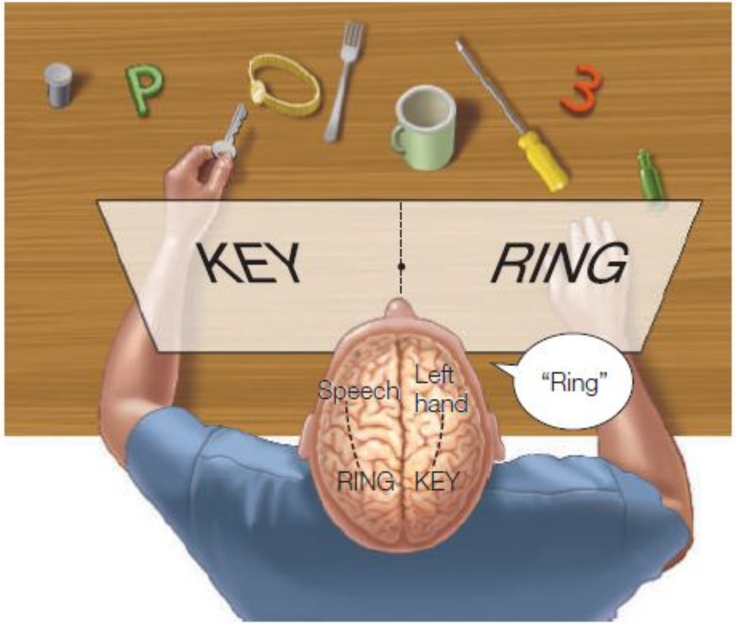


FIGURE 4.11 Restricting visual stimuli to one hemisphere. The split-brain patient reports through the speaking hemisphere only the items flashed to the right half of the screen and denies seeing left-field stimuli or recognizing objects presented to the left hand. Nevertheless, the left hand correctly retrieves objects presented in the left visual field, about which the patient verbally denies knowing anything.

Language and speech

Language processing is preferentially biased to the left hemisphere.

Language and speech are rarely present in both hemispheres; they are in either one or the other (Left in 95–99 percent of right-handed individuals and 70 percent of left-handed).

While the separated left hemisphere normally comprehends all aspects of language, the right hemisphere does have linguistic capabilities, although they are uncommon.

The extent of right-hemisphere language functions is severely limited and restricted to the lexical aspects of comprehension.

Generative syntax is present in only one hemisphere.

Left hemisphere is the dominant hemisphere for speech production in most (96%) of us.

The nonlinguistic, emotional component of speech is called emotional prosody.

Right hemisphere is specialized for comprehending emotional expressions of speech.

The language ability of the left hemisphere perhaps gives it a superior ability to perform higher cognitive functions like making inferences and solving mathematics problems.

Left hemisphere: The interpreter

A hallmark of human intelligence is our ability to make causal interpretations about the world around us.

Causal inferences and interpretations appear to be a specialized ability of the left hemisphere.

After a callosotomy, the verbal intelligence and problem-solving skills of a split-brain patient remain relatively intact. But only in the left hemisphere.

Large part of the right hemisphere’s impoverishment can be attributed to the finding that causal inferences and interpretations appear to be a specialized ability of the left hemisphere.

When the split-brain patient P.S. was given a command to stand up in a way that only the right hemisphere could view, P.S. stood up. When the experimenter asked him why he was standing, P.S.'s speaking left hemisphere immediately came up with a plausible explanation: “Oh, I felt like getting a Coke.”

(If his corpus callosum had been intact, then P.S. would have responded with the right answer).

One constant finding throughout the testing of split-brain patients is that the left hemisphere never admits ignorance about the behavior of the right hemisphere. It always makes up a story to fit the behavior.

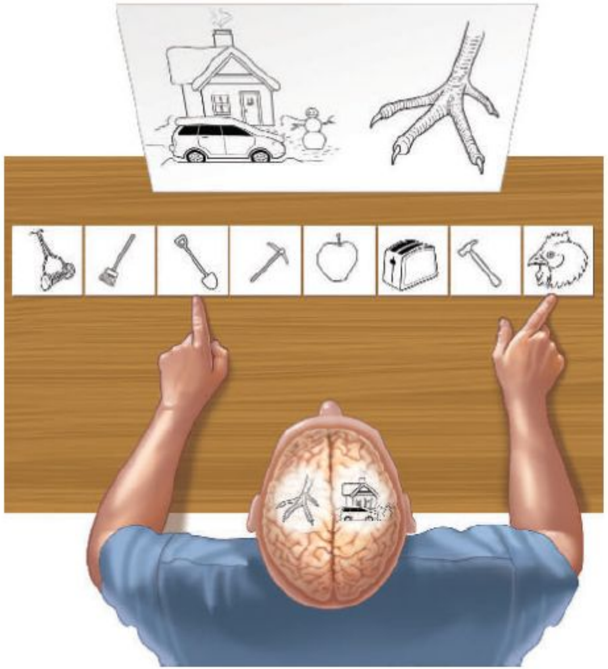


FIGURE 4.29 The interpreter at work.
The left hemisphere of split-brain patient P.S. was shown a chicken claw, and his right hemisphere was shown a snow scene. When P.S. was asked to point to a picture associated with the image he had just seen, his right hand (guided by his left hemisphere) pointed to the chicken, and his left hand pointed to the shovel. When asked why he had pointed to those things, he replied, "Oh, that's simple. The chicken claw goes with the chicken, and you need a shovel to clean out the chicken shed."

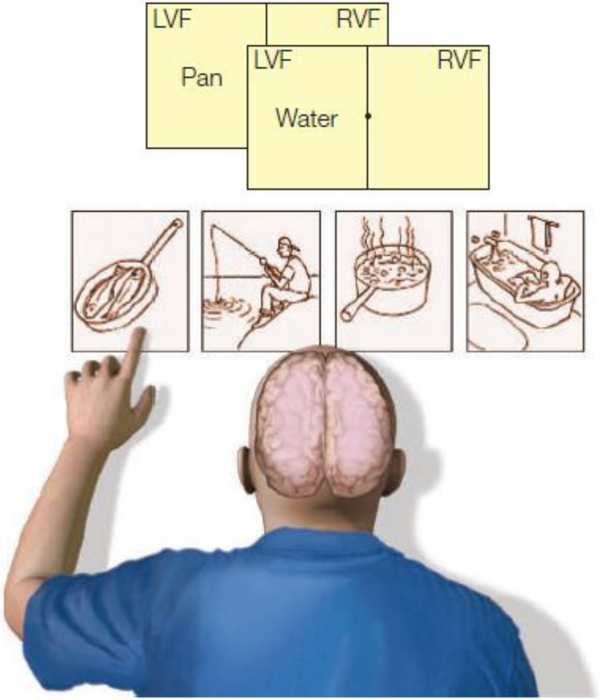


FIGURE 4.31 The right hemisphere is unable to infer causality. The capacity of the right hemisphere to make inferences is extremely limited. In this test, two words are presented in serial order, and the right hemisphere (left hand) is simply required to point to a picture that best depicts what happens when the words are causally related. The left hemisphere finds this sort of task trivial, but the right cannot perform the task.

The unique specialization of the left hemisphere— the interpreter—allows the mind to seek explanations for internal and external events in order to produce appropriate response behaviors.

For the interpreter, facts are helpful but not necessary.

As the interpreter searches for causes and effects, it tries to create order from the chaos of inputs that bombard it all day long.

Global and Local Processing

Time required to identify the global letter was independent of the identity of constituent elements.

While identifying the small letters RT was slowed if the global shape was incongruent with local shape.

Perceptual system first extracts the global shape. Perhaps different processing systems are used for global vs local representations.

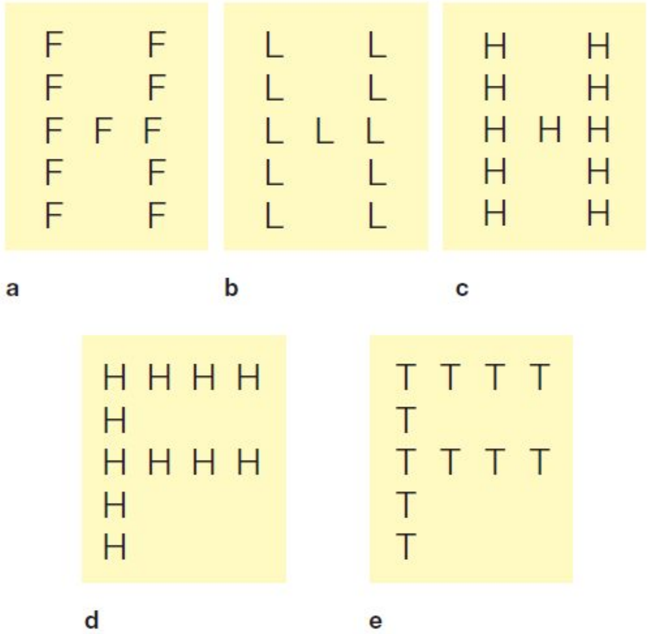


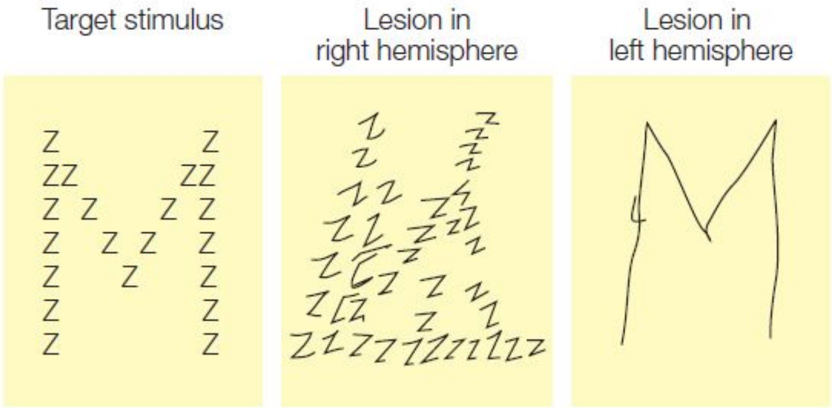
FIGURE 4.26 Local and global stimuli used to investigate hierarchical representation.
Each stimulus is composed of a series of identical letters whose global arrangement forms a larger letter. The participants' task is to indicate whether the stimulus contains an *H* or an *L*. When the stimulus set included competing targets at the two levels (**b**), the participants were instructed to respond either to local targets only or to global targets only. Neither target is present in (**e**).

Patients who had a lesion in either the left or the right hemisphere with local and global stimuli in the center of view.

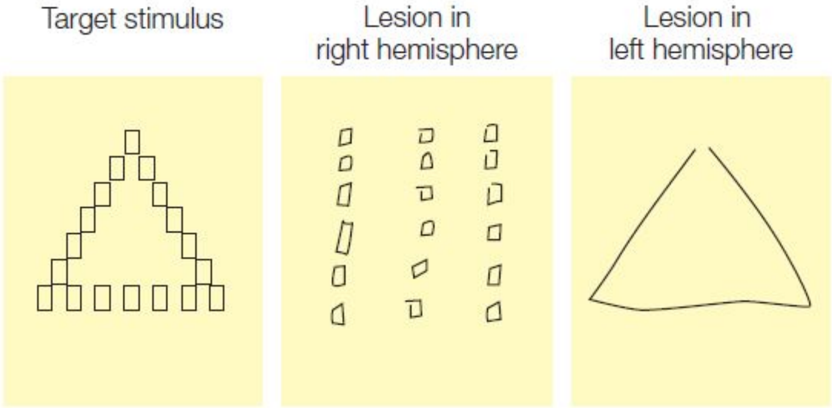
Patients with left -side lesions were slow to identify local targets, and patients with right-side lesions were slow with global targets, demonstrating that the left hemisphere is more adept at representing local information and the right hemisphere is better with global information.

Both hemispheres can abstract either level of representation, but they differ in how efficiently local and global information are represented.

The right is better at the big picture, and the left is more detail oriented.



a Linguistic stimulus



b Nonlinguistic stimulus

Core limbic structures: Hypothalamus

To keep track of the body's internal state, the hypothalamus draws on the same **visceral sensory input pathways**.

Hypothalamus also has sensors that monitor the **composition of the bloodstream** itself.

The hypothalamus can sense concentration of electrolytes, glucose, insulin, sex hormones, metabolic hormones like thyroid-stimulating hormone and growth hormone, markers of infection and immune system activity.

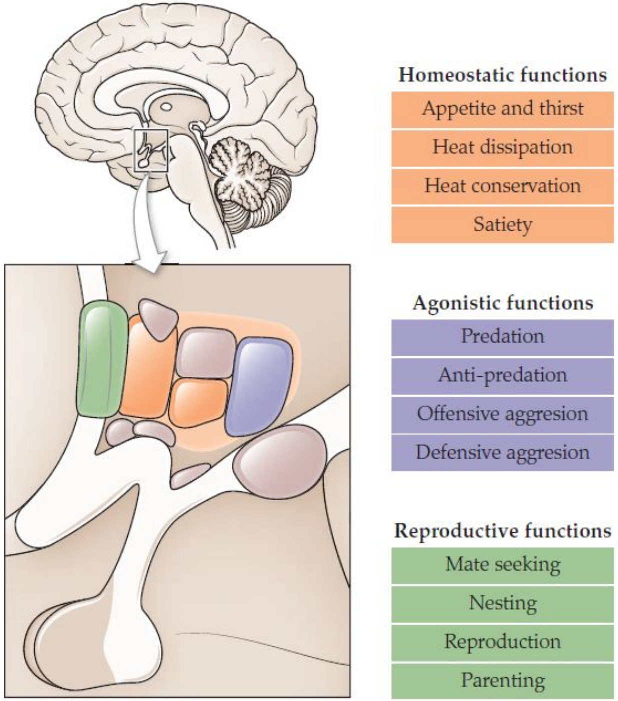


FIGURE 13.10 **The hypothalamus and survival drives.** The nuclei of the hypothalamus coordinate core survival functions, including homeostatic drives, agonistic drives, and reproductive drives.

These kinds of internal sensory inputs are critical for guiding the **appetitive categories of behavior: foraging, feeding, drinking, staying warm**, and so on.

Lateral hypothalamus regulates **hunger** and is counterbalanced by a **ventromedial hypothalamus** that regulates **satiety**.

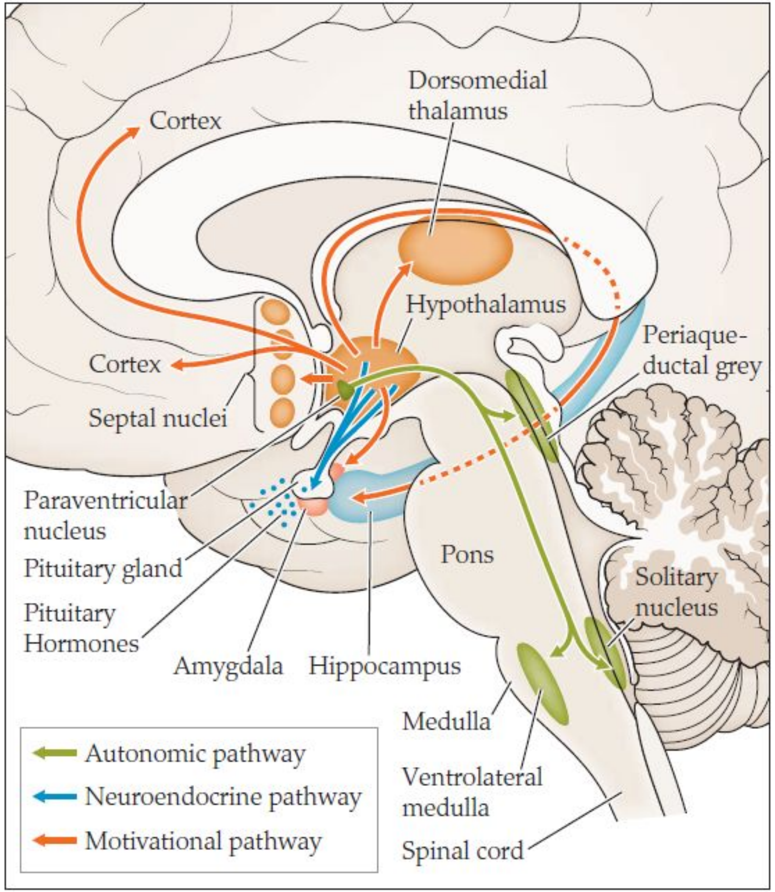
These also play a role in regulating **reproductive functions**, including sexual behavior and parenting behavior.

Agonistic functions also fall under the purview of the hypothalamus. Both **aggressive and defensive behaviors** can be elicited by stimulation of various hypothalamic nuclei.

To effect changes in one’s internal state, the hypothalamus has **three output pathways** available.

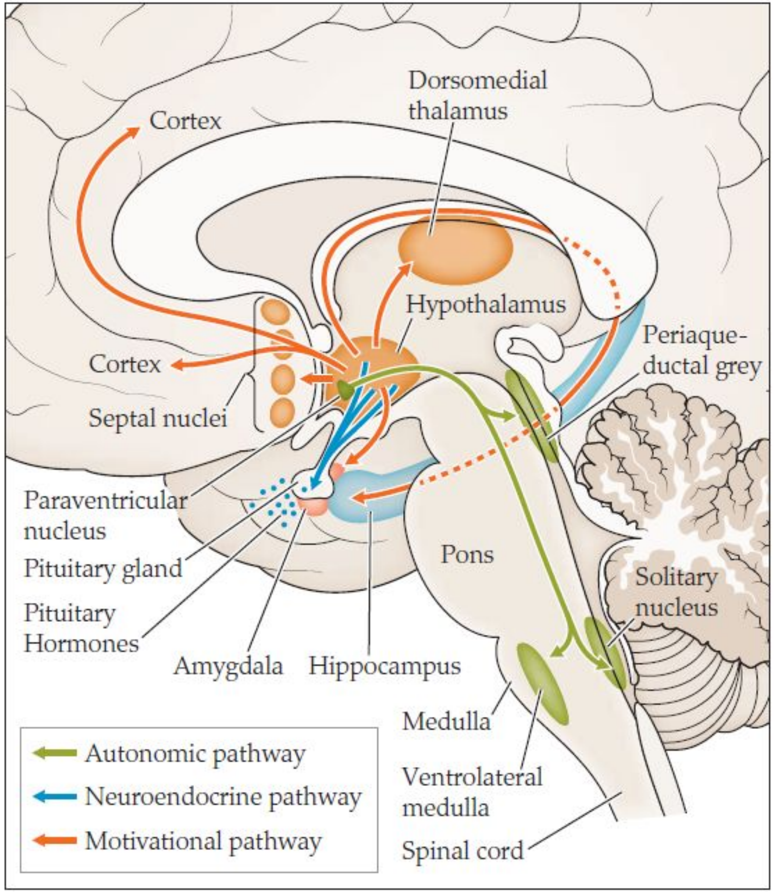
Autonomic output pathway via the **paraventricular nucleus** stimulates the sympathetic and parasympathetic nervous system.

For example, low glucose can prioritize feeding activities like salivation, biting, chewing, and swallowing (simple brainstem action plans).



Neuroendocrine pathway, regulates hormone levels.

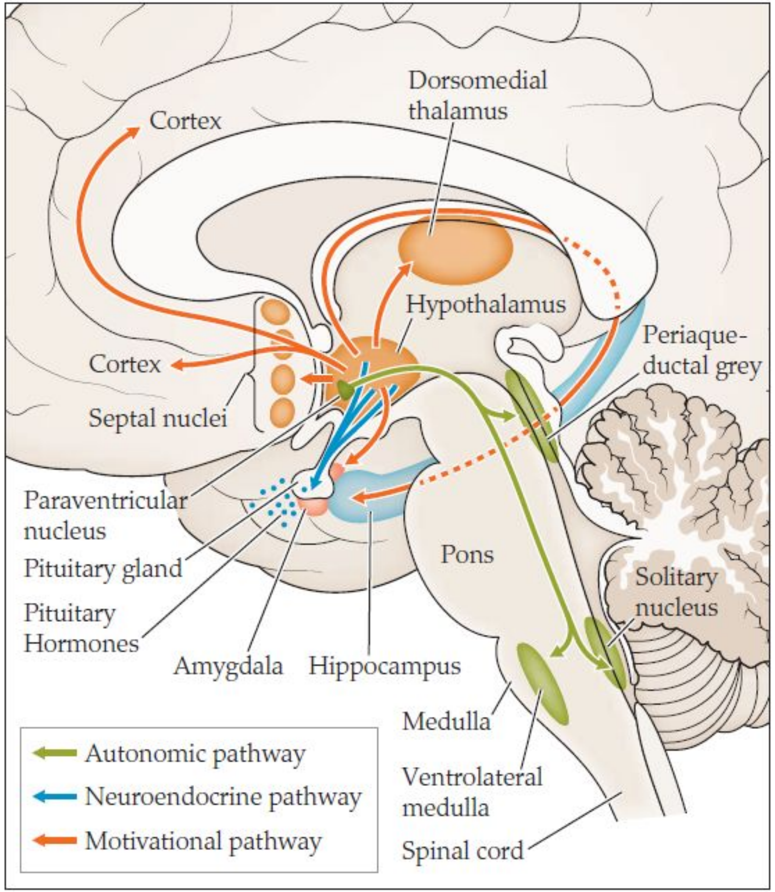
For example, in response **stress**, the hypothalamus secretes **corticotropin-releasing hormone** → **pituitary** → **adrenocorticotropic hormone** (ACTH) → **adrenal glands** → **cortisol**.



The third **pathway is motivational**.

This involves evaluation, goal-setting, strategy formation, behavior selection, action planning, and the matching of actions to the surrounding environment.

This pathway leads to forebrain structures like the striatum and cerebral cortex: areas capable of putting together complex, flexible, nuanced behaviors.



Core limbic structures: Amygdala

Sensory sources of amygdala come from **visual inputs, auditory inputs, and olfactory inputs via the thalamus and cortex**.

Some **interoceptive information** is also available via the ascending pathways through the brainstem.

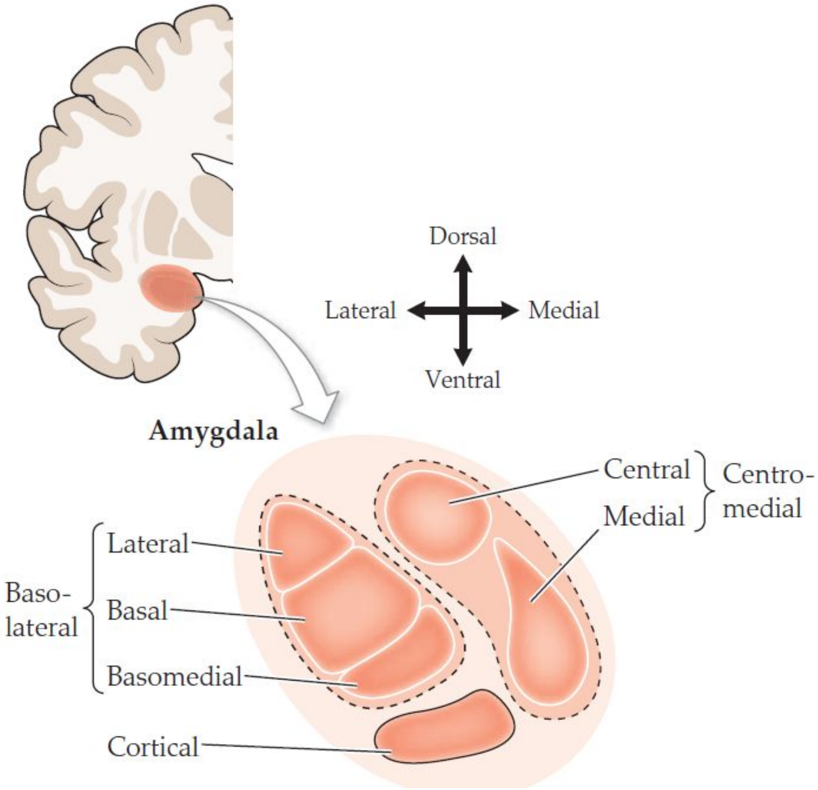
Output pathways lead in three main directions:

- to the **somatic and autonomic nuclei** in the brainstem,
- to the **hypothalamic nuclei** responsible for hormone secretion and for the three main classes of motivated behaviors,
- to the **cortex** involved in evaluation, motivation, and behavioral control.

Basolateral amygdala (input side) has its strongest connections with higher sensory regions in the cortex.

Acts as a **tracker of value** because its neurons are capable of learning and tracking the emotional significance of stimuli from the outside world.

Centromedial amygdala (output side) draws on external-world sensory information to modulate the internal drives and hormonal functions of the hypothalamus.



Key emotional roles for the amygdala lies in the detection of threats and the implementation of fear responses.

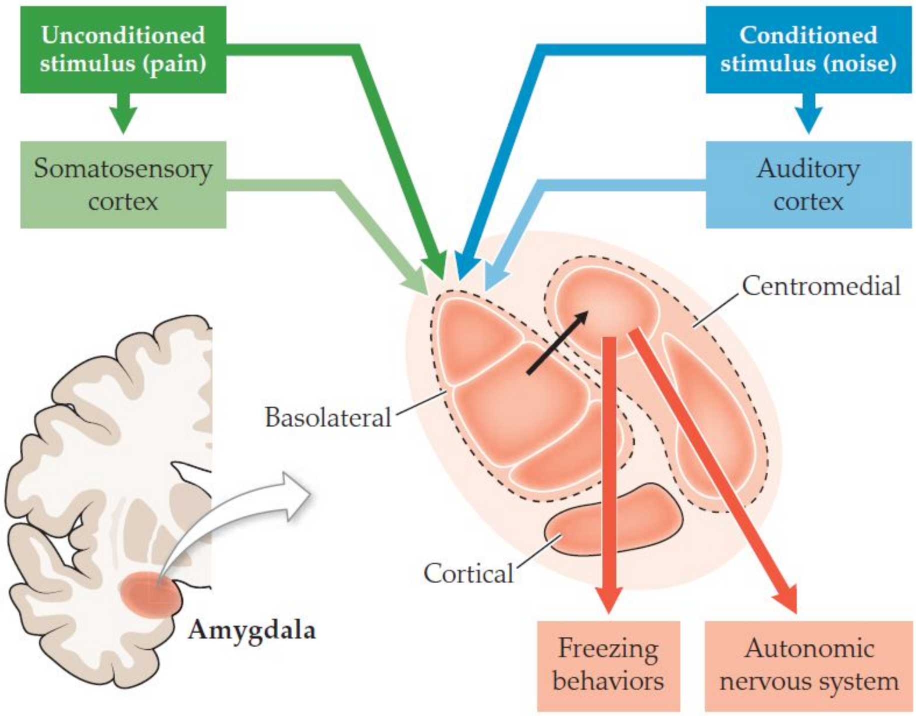


FIGURE 13.16 The classical conditioning pathways in the amygdala. An unconditioned stimulus (US), pain, and a conditioned stimulus (CS), a noise, are presented together. Both the US and the CS stimulate the basolateral amygdala and are paired together, stimulating the central nucleus of the amygdala. The output from the central nucleus triggers the freezing behaviors and activation of the autonomic nervous system. Over a few trials of learning, the connections between these nuclei are strengthened, so that the noise comes to trigger the response on its own.

Some amygdala neurons track positive value, or reward, rather than negative value, or aversiveness.

They can even adjust the reward value based on the organism’s internal state.

So amygdala neurons can take into account the organism’s needs when calculating the value of a stimulus in the outside environment.

Amygdala lesions leads to severe impairments in the learning and expression of fear.

Limbic structures: Hippocampus

The **hippocampus** plays an important role in **spatial navigation** and **episodic memory**: memory for past personal experiences that occurred at a specific time and place, as opposed to memory for facts.

Hippocampus has close connections with amygdala, which helps attaching emotional significance to places or events.

Place cells: Specialised neurons that fire only when the animal is in a particular location within its environment, regardless of the animal's head orientation.

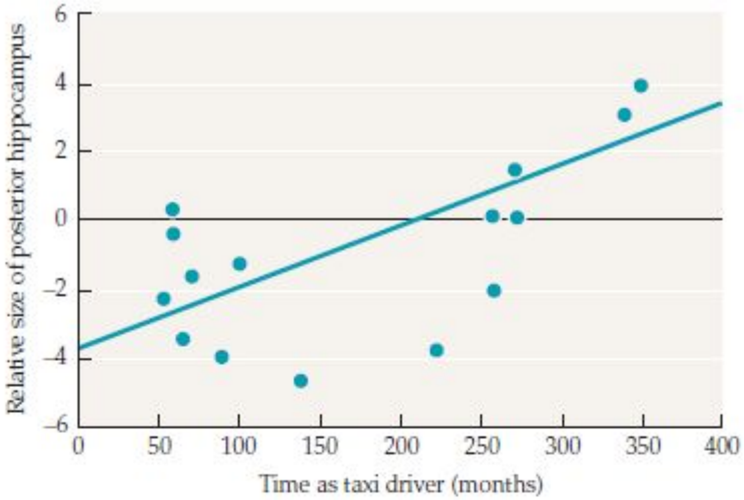


FIGURE 9.10 The posterior hippocampus is larger in taxi drivers who have more months of experience. In a study of London taxi cab drivers, MRI images were collected and used to measure the size of the posterior hippocampus. The relative volume of the posterior hippocampus was greater for subjects who had been driving for a longer time.

Hippocampus and memory

The brain is fundamentally a predictive organ.

Many organisms have a territory, and this territory is filled with potential resources.

Getting to these resources requires a territory map, and this map must be constantly updated with new events to stay accurate.

So spatial maps require episodic memories to be useful for survival.

Hippocampus allows for the rapid learning and high information capacity needed to make the whole system work effectively.